

Analysis of Factors Affecting Match Outcomes in League of Legends

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1 Introduction

The goal of this analysis is to identify which factors have the greatest influence on the outcome of a ranked match in League of Legends, and whether this influence differs depending on the player's position.

I compare two groups of predictors: individual player statistics (KDA ratio, gold per minute, vision score) and team objectives (baron kills, dragon kills, tower kills). I built two logistic regression models to analyze the relationship between these predictors and match outcome, one for each group, and compared them using AUC. I also fit the individual statistics model separately for each position to examine whether the importance of each factor changes depending on the role.

2 Data

The dataset used in this analysis is the *League of Legends Ranked Match Data (2025)*, available on Kaggle [1]. It contains statistics from approximately 68,000 ranked matches played in Season 2025. Each row corresponds to a single player in a single match and includes individual statistics such as kills, deaths, assists, KDA ratio, gold per minute, vision score, as well as team-level statistics like baron kills, dragon kills, tower kills, and the match outcome.

3 Dataset Overview

The dataset contains 68 300 rows across 69 columns. The data covers five positions: TOP, JUNGLE, MIDDLE, BOTTOM and SUPPORT, each represented by roughly 13 600 players. The rank distribution is dominated by Emerald and Platinum players, with fewer players at the extremes (Iron, Challenger).

Looking at the basic statistics, the average match duration is around 30 minutes, with some matches as short as 5 minutes and others exceeding 58 minutes. KDA ratio ranges from 0 to 44, with a median of 2.3, and vision score ranges from 0 to 191.

The win column is perfectly balanced, exactly 50% wins and 50% losses, which is expected since every match has one winning and one losing team.

4 Preprocessing

Before modeling, I performed several cleaning and transformation steps on the raw data.

First, I removed the three rows with position labeled as `Invalid`, as these could not be assigned to any role. Players without a solo rank (674 rows) were assigned the label `UNRANKED` instead of being removed, since they are valid players who simply have not played ranked games yet.

The win column was converted from boolean (`TRUE/FALSE`) to binary (1/0) to make it suitable for logistic regression.

Since each row represents a single player, I aggregated the data to the team level, each team in a match becomes one row. I grouped by `game_id` and `win`, and computed the team average for individual statistics (KDA ratio, gold per minute, damage per minute, vision score) and took the first value for team-level statistics (baron kills, dragon kills,

tower kills, duration), since these are identical for all players on the same team. This resulted in 13,660 rows, two teams per match.

5 Results

5.1 Correlation Matrix

Before fitting any models, I examined the pairwise correlations between all variables of interest. The strongest correlations with the match outcome are `avg_gold_per_min` (0.79) and `tower_kills` (0.77), followed by `avg_kda` (0.72). On the other hand, `avg_vision_score` has almost no correlation with the outcome (0.09).

I also noticed strong correlations between some predictors, most notably `tower_kills` and `avg_gold_per_min` (0.82), which suggests multicollinearity. This will be relevant when interpreting the regression coefficients later.

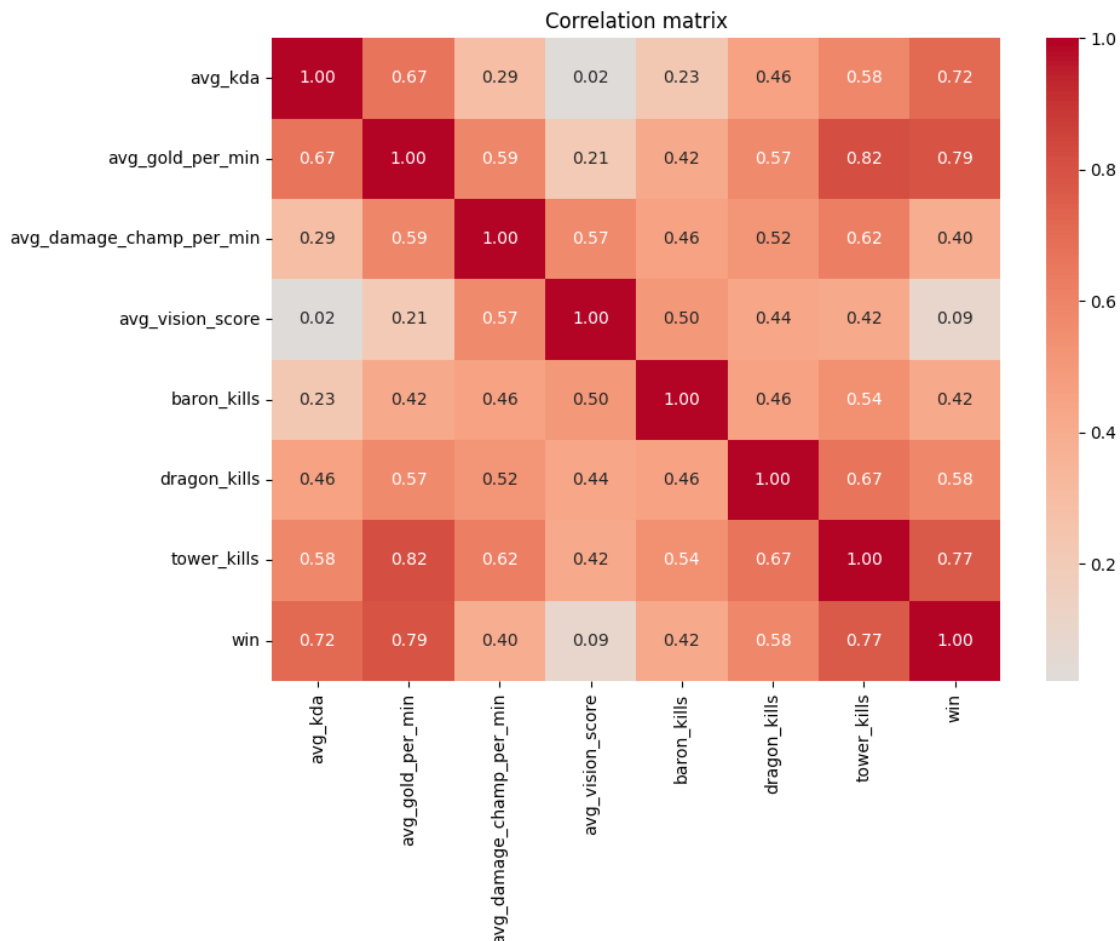


Figure 1: Correlation matrix of team-level variables.

5.2 Boxplots

The boxplots confirm what the correlation matrix suggested, `avg_kda`, `avg_gold_per_min`, `tower_kills` and `dragon_kills` all show a clear difference between winning and losing

teams. The `avg_vision_score` distributions overlap almost completely, consistent with its near-zero correlation with the outcome.

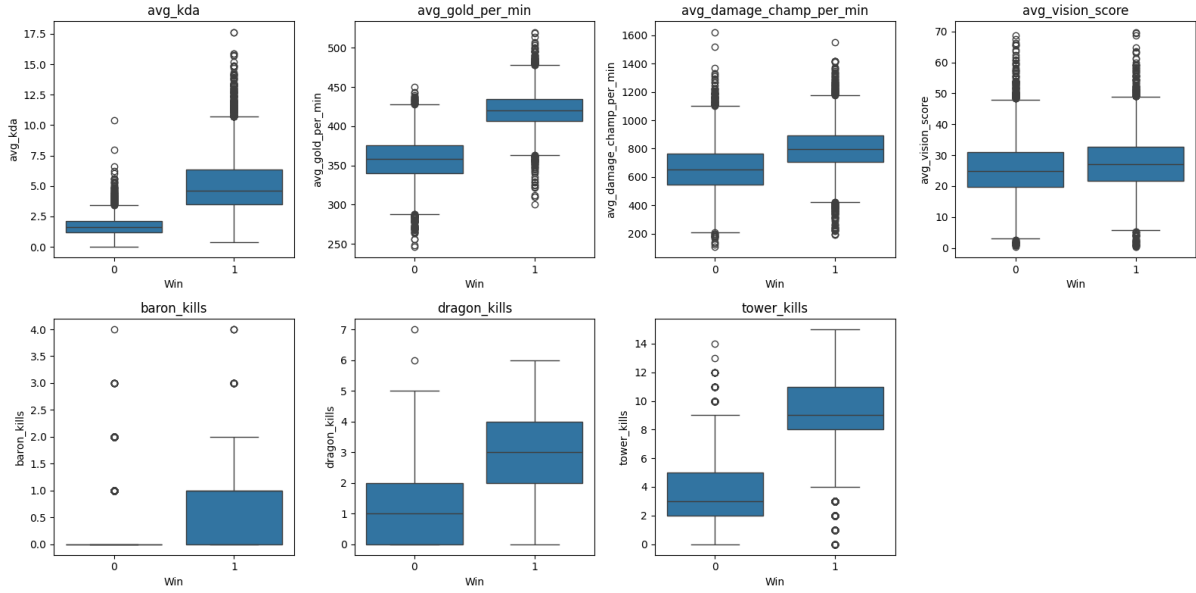


Figure 2: Distribution of each variable by match outcome.

5.3 Logistic Regression Models

I built two logistic regression models to predict the match outcome. Model 1 uses individual player statistics (KDA ratio, gold per minute, vision score) and Model 2 uses team objectives (baron kills, dragon kills, tower kills).

Both models converged successfully. Model 1 achieves an AUC of 0.989 and Model 2 achieves an AUC of 0.940. Since the models are fitted on the full dataset without a train/test split, AUC here reflects how strongly the predictors are associated with the outcome rather than actual predictive performance. Either way, individual statistics are more closely tied to winning than team objectives.

All coefficients in both models are statistically significant ($p < 0.05$). In Model 1, KDA ratio and gold per minute have positive coefficients as expected, better individual performance is associated with a higher probability of winning. Vision score has a negative coefficient, which contradicts its positive correlation with winning (winning teams average 27.4 vs 25.6 for losing teams). This is likely caused by multicollinearity with the other predictors and is discussed further in Section 6.

In Model 2, tower kills and dragon kills have positive coefficients, which makes intuitive sense. Baron kills however has a negative coefficient despite winning teams killing baron roughly four times more often on average. Again, this is an effect of multicollinearity with tower kills.

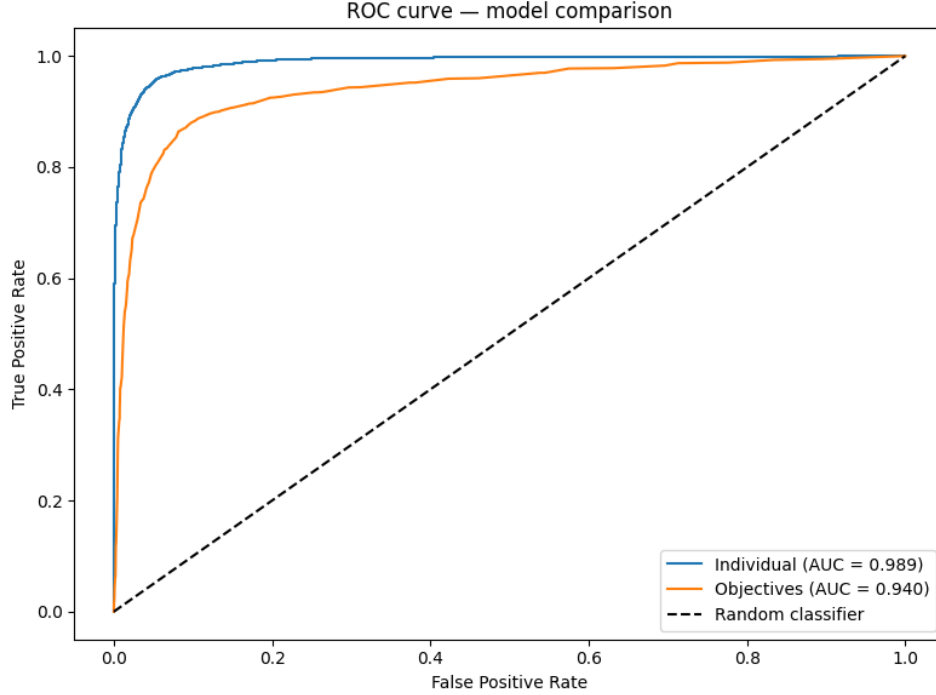


Figure 3: ROC curves for both models.

5.4 Analysis by Position

To examine whether the importance of individual statistics differs by role, I fitted the individual statistics model separately for each position. The results are summarized in the table below.

Position	AUC	KDA coef	Gold coef	Vision coef
TOP	0.866	0.806	0.011	-0.011
JUNGLE	0.909	1.028	0.007	0.006
MIDDLE	0.867	0.791	0.009	-0.005
BOTTOM	0.874	0.899	0.008	-0.007
SUPPORT	0.896	0.656	0.021	0.003

Table 1: Logistic regression coefficients and AUC by position.

The JUNGLE position has the highest AUC (0.909) and the highest KDA coefficient (1.028), suggesting that a jungler’s individual performance has the strongest influence on match outcome among all roles. SUPPORT has the second highest AUC (0.896) and by far the highest gold coefficient (0.021), which makes sense, supports earn less gold overall, so each unit of gold per minute carries more weight.

Vision score is only statistically significant for JUNGLE and SUPPORT, where it has a positive effect. For MIDDLE, the coefficient is not significant ($p = 0.070$). For TOP and BOTTOM it is negative, likely again due to multicollinearity.

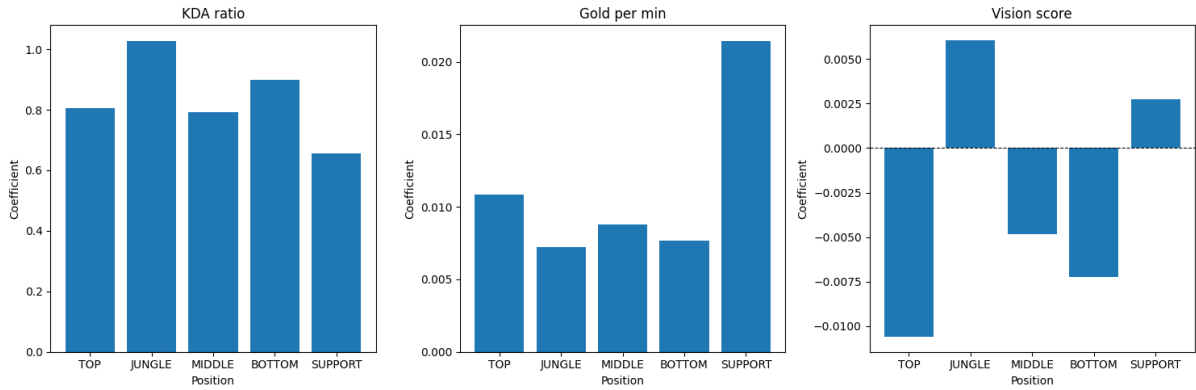


Figure 4: Logistic regression coefficients by position.

6 Discussion

Both models perform well, with AUC values of 0.989 and 0.940, which are high for logistic regression. This suggests a strong statistical association between in-game statistics and match outcomes, leaving relatively little room for randomness.

The biggest problem I ran into was multicollinearity. Several predictors are strongly correlated with each other, for example, tower kills and gold per minute have a correlation of 0.82. This causes some coefficients to have unexpected signs in the models, such as the negative coefficients for vision score and baron kills. However, looking at the univariate averages confirms that these variables do have a positive relationship with winning, winning teams have higher vision scores (27.4 vs 25.6) and kill baron roughly four times more often. The negative coefficients are therefore an artifact of the multicollinearity rather than a real effect.

Another thing worth mentioning is that some combinations of predictor values almost perfectly determine the outcome. For example, a team with a very high number of tower kills will nearly always win. This makes the model overconfident in those cases and contributes to the high AUC values.

The analysis by position uses per-player data rather than aggregated team data, so players from the same match are not fully independent. This is a limitation, but since each position is analyzed separately and the dataset is fairly large, I think the results are still reasonable.

7 Conclusion

In this analysis I tried to find out which factors influence the outcome of a ranked League of Legends match using logistic regression. I compared two models, one based on individual player statistics and one based on team objectives, and found that individual statistics (KDA ratio, gold per minute, vision score) are more strongly associated with match outcomes than team objectives (AUC 0.989 vs 0.940).

Looking at individual positions, the jungler's performance has the strongest influence on match outcome, with the highest KDA coefficient and the highest AUC among all roles. Support has by far the highest gold per minute coefficient among all positions,

which reflects the nature of the role – supports earn less gold overall, so each unit of gold per minute carries more weight.

To sum up, individual player performance seems to matter more than team objectives when it comes to predicting match outcomes, though objectives are also relevant. The high AUC values suggest that what happens in a LoL match is pretty well captured by the stats alone.

References

- [1] Jakub Krasuski, *League of Legends Ranked Match Data (2025)*, Kaggle, 2025. <https://www.kaggle.com/datasets/jakubkrasuski/league-of-legends-ranked-match-data-season-15>